



DEPARTMENT OF ENGINEERING CYBERNETICS

TTK4900

MASTER THESIS

Adaptive Filtering of Blood Velocity Signals During Running

Author:

Anders Haarberg Eriksen

Date: 11th September 2026

Abstract

Acknowledgments

Contents

1	Introduction	1
2	Theoretical background	2
2.1	Ultrasound	2
2.1.1	Transducer	2
2.1.2	Reflection and Scattering	2
2.1.3	Attenuation	2
2.1.4	Time of Flight	2
2.1.5	Noise sources	2
2.2	Pulsed-Wave Doppler	2
2.2.1	Decimation	3
2.2.2	Clutter Filter	3
2.3	IMU	3
2.3.1	Sensor Fusion	3
2.4	The Cardiovascular system	4
2.4.1	Artieries an Veins	4
2.4.2	NOTES	4
3	Method	6
4	Results	7
5	Discussion	8
6	Conclusion	9

List of Figures

2.1	Ideal emitted electrical waveform for Pulsed-Wave Doppler (PW Doppler), with annotations. Note that the time axis is broken to show the pulse repetition period.	3
-----	--	---

List of Tables

Chapter 1

Introduction

Chapter 2

Theoretical background

2.1 Ultrasound

Ultrasound is defined as sound waves with frequencies above the upper limit of human hearing, which is typically around 20 kHz.

2.1.1 Transducer

2.1.2 Reflection and Scattering

2.1.3 Attenuation

2.1.4 Time of Flight

2.1.5 Noise sources

2.2 Pulsed-Wave Doppler

In the present system, PW Doppler is used to measure the blood flow velocity. It works by emitting a series of high frequency sound pulses into the tissue at a constant Pulse Repetition Frequency (PRF). Each pulse consists of a burst of sound waves with a pulse center frequency f_c with N_{cycles} cycles. The pulse length T is then given by $T = N_{cycles}/f_c$. After a pulse is emitted, the system records the returning echoes. This is commonly referred to as the *fast time* signal. The fast time signal is sampled with a sampling rate $f_{s,FT}$

Parameter	Symbol	Value
Pulse center frequency	f_c	8 MHz
Number of cycles	N_{cycles}	12
Pulse length	T	1.5 μs
Pulse repetition frequency	PRF	20 kHz
Speed of sound in tissue	c	1540 m s^{-1}

Ideal Electrical Waveform for PW Doppler

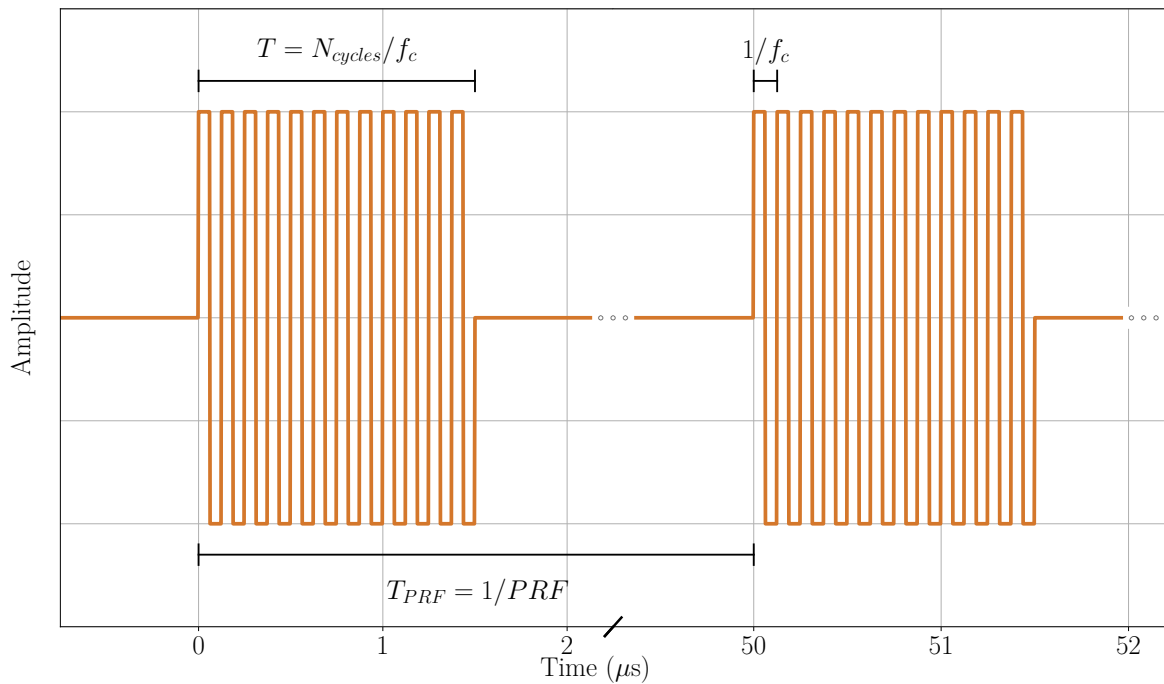


Figure 2.1: Ideal emitted electrical waveform for PW Doppler, with annotations. Note that the time axis is broken to show the pulse repetition period.

2.2.1 Decimation

2.2.2 Clutter Filter

2.3 IMU

2.3.1 Sensor Fusion

Considering the gyro on the Inertial Measurement Unit (IMU) is prone to drift over time, sensor fusion algorithms are employed to combine the measurements from the accelerometer and the gyroscope to obtain a more accurate estimate of the orientation. In our IMU we have access to linear acceleration from the accelerometer, and angular velocity from the gyroscope. There are several ways to perform sensor fusion, including complementary filters, Kalman filters, and more advanced nonlinear filtering techniques.

Complementary Filter

The idea behind the complementary filter is to

2.4 The Cardiovascular system

2.4.1 Arteries and Veins

TODO: Add details on which arteries and veins are important in this thesis.

2.4.2 NOTES

TODO: Add details on how the heart functions

During *systole*, the ventricles contract, ejecting blood out of the heart and into the arteries. This phase is followed by *diastole*, during which the ventricles relax and fill with blood from the atria. This is a cyclic process that repeats with each heartbeat. The two phases can be observed in the blood flow velocity as a systolic peak during ventricular contraction and a diastolic peak during ventricular relaxation. **TODO: Am I correct in this?**

The pressure wave emitted by the heart mostly remains unchanged in large and mid-size arteries, like the radial artery. Only a drop in mean pressure of a few mmHg can be observed, due to viscous effects.

The peak value of the pressure, measured at any location, is called the *systolic pressure* p_s . The minimal pressure value is denoted the *diastolic pressure* p_d . The difference between the systolic and diastolic pressures is known as the *pulse pressure* $p_{pulse} = p_s - p_d$. Blood pressure is traditionally expressed in mmHg (1 mmHg $\approx$ 133.322 Pa).

Compliance describes the change in vessel dimensions in response to a change in internal pressure. Hellevik (2018) defines compliance as a local property of the vessel wall and expresses it in terms of the cross-sectional area A and the internal pressure p as:

$$C = \frac{\partial A}{\partial p}. \quad (2.1)$$

This represents the change in cross-sectional area per unit change in pressure and can be interpreted as a local measure of vessel compliance. A related quantity is the *distensibility* D , which is the compliance normalized by the cross-sectional area,

$$D = \frac{1}{A} \frac{\partial A}{\partial p} = \frac{C}{A}. \quad (2.2)$$

Blood vessels with a wall thickness h much smaller than the vessel radius r , such that $h \ll r$, are referred to as *thin-walled vessels*. For thin-walled vessels, the compliance can be approximated as:

$$C \approx 2 \frac{r}{h} \frac{A}{\eta}, \quad (2.3)$$

where η is the Young's modulus of the vessel wall material (**POSSIBLE TODO: ADD PROOF OF THIS**). As a general rule of thumb, blood vessels are considered thin-walled when the ratio of the radius to the wall thickness, $\frac{r}{h}$, is greater than 10 (Ibrahim et al., [2015](#)). Getachew et al. ([2018](#)) measured did a study on the radial artery in

Chapter 3

Method

Chapter 4

Results

Chapter 5

Discussion

Chapter 6

Conclusion

Bibliography

- Getachew, D., Astatkie, A., & Lemma, K. (2018). Diameter, vessel thickness and angle of bifurcation of the radial artery in ethiopian cadavers. *Journal of Morphological Sciences*, 35(2), 129–135. <https://doi.org/10.1055/s-0038-1669905>
- Hellevik, L. R. (2018). Cardiovascular biomechanics.
- Ibrahim, A., Ryu, Y., & Saidpour, M. (2015). Stress analysis of thin-walled pressure vessels. *Modern Mechanical Engineering*, 05(1), 1. <https://doi.org/10.4236/mme.2015.51001>